Module 3 — Pythia 8 Advanced

Week 3 · Hard QCD, tunes, and LHAPDF

3.1 Hard vs. soft processes

Soft processes are dominated by low-momentum-transfer QCD, which cannot be computed perturbatively and is modeled phenomenologically (Regge phenomenology, MPI, string fragmentation). **Hard** processes have a large momentum transfer Q^2, so their cross section factorizes into a perturbatively calculable matrix element times parton distribution functions (PDFs).

In Pythia, switch from SoftQCD to HardQCD:

```
Beams:eCM = 200.
HardQCD:all = on
PhaseSpace:pTHatMin = 10.
PhaseSpace:pTHatMax = 100.
Tune:pp = 14
```

Subtle: pTHatMin is the minimum parton-level pT of the hard scattering. It is NOT the same as final-state jet pT. Showering and hadronization broaden the distribution.

3.2 Tunes

A **tune** is a consistent set of values for parameters governing MPI, the shower, color reconnection, and hadronization. Tunes are derived by fitting to data. Common choices: Monash 2013 (default, Tune:pp = 14), CUETP8M1 (CMS), A14 (ATLAS). For sPHENIX at $\sqrt{s}$=200 GeV, Monash is a reasonable default; the sPHENIX collaboration also maintains an in-house RHIC-tuned set.

3.3 PDFs via LHAPDF (optional but recommended)

By default Pythia ships with its own built-in PDF set (NNPDF2.3 LO). For serious work install LHAPDF6 and select a modern set such as NNPDF3.1 NLO:

```
PDF:pSet = LHAPDF6:NNPDF31_nlo_as_0118
```

Install LHAPDF with:

```
wget https://lhapdf.hepforge.org/downloads/?f=LHAPDF-6.5.4.tar.gz -O LHAPDF-6.5.4.tar.gz
tar xf LHAPDF-6.5.4.tar.gz && cd LHAPDF-6.5.4
./configure --prefix=$HOME/sw/lhapdf && make -j$(nproc) install
lhapdf install NNPDF31_nlo_as_0118
./configure --prefix=$HOME/sw/pythia8 --with-lhapdf6=$HOME/sw/lhapdf \
--with-hepmc3=$HEPMC3_DIR
make -j$(nproc) install # rebuild Pythia
```

3.4 Building a jet sample

Use FastJet for anti-kT clustering. Install FastJet (fastjet.fr), then modify the Module 2 code:

```
#include "fastjet/ClusterSequence.hh"
using namespace fastjet;
```

```
std::vector<PseudoJet> fj;
for (int i=0; i<pythia.event.size(); ++i) {
auto& pp = pythia.event[i];
if (!pp.isFinal()) continue;
if (fabs(pp.eta()) > 1.5) continue;
fj.emplace_back(pp.px(), pp.py(), pp.pz(), pp.e());
}
JetDefinition jetDef(antikt_algorithm, 0.4);
ClusterSequence cs(fj, jetDef);
auto jets = sorted_by_pt(cs.inclusive_jets(5.0));
```

3.5 Writing HepMC3 output

To hand events to Geant4 or to another program, write HepMC3 ASCII:

```
#include "Pythia8Plugins/HepMC3.h"
Pythia8::Pythia8ToHepMC toHepMC("events.hepmc");
for (int iEv=0; iEv<N; ++iEv) {
if (!pythia.next()) continue;
toHepMC.writeNextEvent(pythia);
}
```

3.6 Exercises

- Generate 50k HardQCD events with pTHatMin=10 GeV at $\sqrt{s}$=200 GeV. Plot the leading-jet pT spectrum.
- Write the sample to HepMC3 and verify with HepMC3-config --version & a quick head.
- Toggle between Monash and CUETP8M1; compare charged-particle density $dN/d\eta$.

3.7 Self-assessment

- [ ] I understand the distinction between pTHat and jet pT.
- [ ] I can swap PDF sets via LHAPDF.
- [ ] I produced a HepMC3 file readable by another tool.